

Transcriptomic profiling of dwarf tomato (*Solanum lycopersicum* cv. ‘Red Robin’) in spaceflight: evaluating the main effects of microgravity under red- and blue-light covariates

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Abstract

Cultivating nutrient-dense crops in microgravity is a prerequisite for long-duration human space exploration, yet the molecular adaptations of fruiting crops to spaceflight remain incompletely resolved. Using RNA-sequencing data from the NASA VEG-05 experiment (Open Science Data Repository OSD-767 / GLDS-709), we profiled leaf ($n = 21$) and adventitious-root ($n = 15$) tissues of dwarf tomato grown aboard the International Space Station under red- and blue-rich light versus ground controls. Modelling light spectrum as a covariate ($\sim \text{light} + \text{condition}$) to isolate the main effect of spaceflight, we identified 2,132 differentially expressed genes (DEGs) in leaf (1,032 up / 1,100 down) and 2,582 in adventitious root (1,706 up / 876 down), with roots showing a larger and more strongly up-regulated response. Functional enrichment converged on oxidative-stress management and specialised metabolism: root up-regulated genes were dominated by “response to oxidative stress” (GO, $\text{FDR} \approx 6.5 \times 10^{-13}$), hydrogen-peroxide catabolism and ethylene signalling, and by KEGG phenylpropanoid ($\text{FDR} \approx 4 \times 10^{-30}$), flavonoid and stilbenoid biosynthesis with glutathione metabolism. The plant circadian-rhythm pathway was down-regulated in both organs. Cross-referencing root DEGs against a published tomato root cell-type atlas localised the response to cortex, exodermis, endodermis, meristematic-cortex and vascular cell populations. These light-averaged responses, which independently corroborate the original VEG-05 report, identify antioxidant and phenylpropanoid metabolism as core targets for engineering spaceflight-resilient crops.

Introduction

Long-duration space missions require bioregenerative life-support systems that supply fresh, nutrient-dense food. The dwarf tomato (*Solanum lycopersicum* cv. ‘Red Robin’) is a leading candidate for space agriculture owing to its compact habit, continuous fruiting and nutritional value. Understanding how this crop adapts to the coupled stressors of spaceflight—microgravity, altered fluid behaviour and the closed cultivation environment—is essential to optimise off-Earth cultivation.

Plant physiology is strongly shaped by light quality. In the Vegetable Production System (Veggie) aboard the International Space Station (ISS), red and blue wavelengths govern photosynthetic efficiency, phototropism and stress signalling through phytochrome, cryptochrome and phototropin photoreceptors. It remains unclear how the foundational transcriptomic response to microgravity operates across, or interacts with, these light cues. The VEG-05 experiment (OSD-767) was designed to evaluate the impact of red- versus blue-rich light on tomato growth, nutrition and microbial safety in space. Here we leverage its transcriptomic data to isolate the fundamental genetic adaptations of *S. lycopersicum* to spaceflight. Rather than testing a flight-by-light interaction, we model light condition as a covariate, determining the robust pri-

mary effects of spaceflight on leaf and adventitious-root transcriptomes averaged across light spectra, and thereby define the core molecular footprint of spaceflight adaptation in a vital crop species.

Results

Data quality, dispersion and sample ordination

After trimming and alignment, per-gene dispersion estimates conformed to the expected mean-variance relationship in both tissues, supporting the negative-binomial models. Unsupervised ordination of variance-stabilised counts separated samples primarily by spaceflight condition: the leading principal components distinguished Flight from Ground samples, while red- and blue-light replicates interleaved within each condition (Fig. 1), consistent with treating light spectrum as a covariate. The larger within-group spread among root samples reflects the limited and imbalanced root ground-control replication ($n = 3$ per light condition).

Spaceflight extensively remodels the leaf transcriptome

At an adjusted $p < 0.05$ and $|\log_2 \text{FC}| \geq 1$, the leaf Flight-versus-Ground contrast identified **2,132 significant DEGs** (1,032 up, 1,100 down in spaceflight; Table 1). The response was both broad and highly significant: 1,337 genes passed $\text{FDR} < 0.001$. Indi-

vidual effect sizes were moderate, the most strongly induced leaf genes reaching $\log_2 \text{FC} \approx +5.6$ (e.g. *Solyc07g052370*, $\log_2 \text{FC} \approx +5.0$, $\text{FDR} \approx 1 \times 10^{-17}$) and the most repressed ≈ -4.5 . Volcano and MA representations (Fig. 2) show a broadly symmetric distribution of induced and repressed genes across the expression range.

Adventitious roots display a larger, up-regulation-biased response

The adventitious-root contrast identified **2,582 significant DEGs**, exceeding the leaf response in magnitude and asymmetry: 1,706 genes up- versus 876 down-regulated ($\approx 2:1$ induction bias; 638 at $\text{FDR} < 0.001$). Effect sizes were markedly larger than in leaf—numerous root genes were induced from near-absent baseline expression, the strongest exceeding $10 \log_2$ -units (the single most extreme $\approx +25$ from a near-zero ground baseline), indicating switch-like activation of otherwise silent loci in spaceflight. Sample-distance clustering, ordination, volcano and MA plots (Figs. 1, 3) consistently separated Flight from Ground root samples and highlighted the up-regulation bias.

Enrichment implicates oxidative-stress management and specialised metabolism

Gene Ontology (GO) and KEGG over-representation analysis of the directional DEG sets revealed a coherent, stress-centred programme. In **roots**, up-regulated genes were overwhelmingly enriched for oxidative-stress management: the GO biological processes “response to oxidative stress” (32 genes, $\text{FDR} \approx 6.5 \times 10^{-13}$) and “hydrogen peroxide catabolic process” (28 genes) were the leading terms, alongside “ethylene-activated signaling pathway” ($\text{FDR} \approx 1 \times 10^{-6}$) and, at the molecular-function level, an exceptionally strong heme-binding signature (67 genes, $\text{FDR} \approx 9 \times 10^{-18}$) reflecting peroxidase and cytochrome activity (Fig. 4a). The corresponding KEGG analysis was dominated by **phenylpropanoid biosynthesis** (45 genes, $\text{FDR} \approx 4 \times 10^{-30}$), followed by stilbenoid, flavonoid and aromatic-amino-acid biosynthesis and by **glutathione metabolism** (Fig. 4b)—a concerted mobilisation of antioxidant and secondary-metabolite pathways.

In **leaves**, up-regulated genes showed a related but attenuated signature: enrichment for heme binding, monooxygenase/oxidoreductase and iron-ion binding (Fig. 5a), induction of abscisic-acid-binding genes, and KEGG enrichment for phenylpropanoid and flavonoid biosynthesis, amino-acid biosynthesis, MAPK signalling and plant-hormone signal transduction (Fig. 5b). Down-regulated genes pointed to suppression of normal growth and timekeeping: the **plant circadian-rhythm** KEGG pathway was significantly down-regulated in *both* organs, roots additionally down-regulating Calvin-cycle carbon fixation and starch/sucrose metabolism, and leaves down-regulating transmembrane transport, galactose and

folate metabolism and glycolysis (Figs. 4c, 5c).

Root DEGs localise to ground-tissue and vascular cell types

To interpret the bulk response cellularly, root DEGs were cross-referenced against published tomato root cell-type marker sets³ by Fisher’s exact test. Twenty-five cell-type $\times$ DEG associations were significant (Fig. 6). Down-regulated root genes were most strongly enriched in **cortex** (odds ratio ≈ 3.6 , $\text{FDR} \approx 7 \times 10^{-29}$), **endodermis**, **meristematic cortex** and **xylem/vascular** markers, whereas up-regulated genes were most enriched in **exodermis** ($\text{OR} \approx 2.8$) and an actin-defined cortex population (Cortex_ACT2, $\text{OR} \approx 4.4$). The response therefore concentrates in the ground-tissue layers and the vasculature—precisely the cell populations re-specified during adventitious-root formation. Cross-referencing against a leaf cluster atlas gave a weaker, less tissue-specific pattern.

Discussion

This study isolates the core, light-averaged transcriptional response of dwarf tomato to spaceflight in the two organs most relevant to crop performance—photosynthetic leaves and the adventitious roots that proliferate in microgravity. By modelling light spectrum as a covariate we traded sensitivity to light-specific effects for power on the main effect of spaceflight, recovering a pervasive response of $\sim 2,100$ leaf and $\sim 2,600$ root DEGs.

Adventitious roots are the focal organ. Roots contained more DEGs than leaves, showed a pronounced up-regulation bias and far larger effect sizes, including switch-like activation of loci essentially silent in ground controls. This intensity mirrors the documented phenotype of extensive adventitious rooting in spaceflight, and our cell-type cross-reference ties the signal specifically to cortex, exodermis, endodermis and vascular populations—the tissues that physically remodel during rooting. Spaceflight adaptation in tomato roots is thus, in large part, a programme of ground-tissue and vascular re-organisation rather than a uniform whole-organ response.

A shared oxidative-stress and phenylpropanoid signature. The most robust theme across both organs is management of reactive oxygen species. Extreme enrichment of oxidative-stress response, hydrogen-peroxide catabolism and heme/peroxidase activity in roots, with induction of glutathione metabolism, indicates that antioxidant defence is a central axis of adaptation. This couples to strong mobilisation of the phenylpropanoid and flavonoid pathways—versatile sources of antioxidant and lignin-precursor metabolites—the strongest KEGG signal in roots and also significant in leaves. Because phenylpropanoid-derived compounds feed both ROS scavenging and cell-wall reinforcement, their induction plausibly serves the dual demands of oxidative defence and the mechanical remodelling accompanying

adventitious rooting.

Hormonal and circadian reprogramming. Ethylene signalling in roots and abscisic-acid-binding genes in leaves implicate classical stress hormones, reinforced by leaf MAPK and hormone signal-transduction enrichment. Down-regulation of the circadian-rhythm pathway in *both* organs is notable: circadian disruption recurs across spaceflight biology, and its appearance in a light-controlled analysis argues it is a genuine microgravity-associated effect rather than a light artefact. Concurrent suppression of primary carbon and energy metabolism is consistent with resource re-allocation from growth toward defensive and structural metabolism.

Concordance with the original VEG-05 report. These findings independently corroborate the published analysis of this dataset¹, which likewise reported pronounced adventitious-root change and highlighted oxidative-stress response, secondary-metabolite biosynthesis and hormonal regulation. We reach these conclusions through a different statistical route—an additive covariate model estimating the pooled main effect of spaceflight, rather than within-light contrasts intersected across treatments—and consequently report larger DEG counts under a more permissive fold-change threshold. The analyses are complementary: the published within-light design isolates a conservative core of light-invariant genes, whereas the present design maximises power to describe the full breadth of the response and to localise it to specific cell types.

Limitations. The root ground-control group comprises only three replicates per light condition, limiting power and inflating dispersion for root contrasts; formal light×spaceflight interaction testing in root is under-powered (an exploratory interaction analysis is provided in the associated data repository). *Solanum*-to-Entrez identifier mapping recovered 60–75% of genes, so enrichment under-samples unmapped loci. The cell-type cross-reference relies on published markers rather than a spaceflight-specific single-cell atlas, so cell-type assignments are inherited from ground-grown reference tissue.

Conclusions. Averaged across red- and blue-rich light, spaceflight imposes a coordinated, stress-oriented transcriptional programme on dwarf tomato that is strongest in adventitious roots and built around antioxidant defence, phenylpropanoid/flavonoid metabolism, stress-hormone signalling and circadian dysregulation, with the root response localised to ground-tissue and vascular layers. These pathways constitute a concise set of candidate targets—peroxidase/glutathione antioxidant systems and phenylpropanoid flux in particular—for breeding or engineering tomato cultivars better suited to bioregenerative life-support systems.

Methods

Data acquisition and experimental design

Raw paired-end RNA-seq reads (151 bp, NovaSeq X Plus) for VEG-05 were obtained from the NASA Open Science Data Repository (OSD-767; GeneLab GLDS-709). The design comprised 21 leaf samples (Flight/Red $n=5$, Flight/Blue $n=4$, Ground/Red $n=6$, Ground/Blue $n=6$) and 15 adventitious-root samples (Flight/Red $n=5$, Flight/Blue $n=4$, Ground/Red $n=3$, Ground/Blue $n=3$).

Pre-processing and quality control

Processing followed the NASA GeneLab RNAseq Consensus Pipeline (GL-DPPD-7101-G / NF_RCP 2.0.1). Read quality was assessed with FastQC v0.12.1 and aggregated with MultiQC v1.26; adapters and low-quality bases were removed with Trim Galore! v0.6.10 (Cutadapt v4.2; -phred33, paired-end), followed by a second FastQC pass.

Alignment and quantification

The Ensembl Plants *S. lycopersicum* SL3.0 assembly with ITAG gene models was used as reference. Trimmed reads were aligned with STAR⁵ v2.7.11b in two-pass mode (-sjdbOverhang 150, -outFilterMultimapNmax 20, -outFilterMismatchNoverReadLmax 0.04); transcriptome alignments were quantified with RSEM⁶ v1.3.1 (-estimate-rspd). Strandedness and library geometry were verified with RSeQC v5.0.4.

Differential expression

Gene-level counts were imported with tximport v1.38.2 into DESeq2⁴ v1.50.2 (R v4.4.2). Leaf and root tissues were analysed independently. To isolate the primary effect of spaceflight while controlling for light, the design was $\sim \text{light} + \text{condition}$, with the Flight-versus-Ground contrast as primary. Low-count genes were pre-filtered; $\log_2$ fold changes were shrunk for visualisation. DEGs were defined by Benjamini-Hochberg FDR < 0.05 and $|\log_2 \text{FC}| \geq 1$. Volcano, MA and PCA plots were produced with EnhancedVolcano v1.28.2 and DESeq2.

Functional enrichment

GO (BP/MF/CC) and KEGG (organism = "sly") over-representation was performed with clusterProfiler²; Solyc identifiers were mapped to Entrez via biomaRt (Ensembl Plants). Enrichment used relaxed cut-offs for small gene sets; the combined results are provided as a supplementary table.

Cell-type cross-reference

Directional DEG sets were intersected with published tomato root cell-type marker lists³; per-cell-type over-representation was tested by Fisher's exact test with Bonferroni correction and summarised as $-\log_{10}$ adjusted p .

Data availability

Primary data are available from the NASA Open Science Data Repository under accession [OSD-767](#) (GLDS-709). All processed results, figures, code trace and this manuscript are archived in the project repository and will be deposited on Zenodo (DOI to be assigned).

Acknowledgements

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Competing interests

The author declares no competing interests.

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Table 1 | Differential expression summary (Flight vs Ground). DESeq2 additive model $\sim$ light + condition; DEGs at $FDR < 0.05$, $|\log_2 FC| \geq 1$.

Tissue	<i>n</i>	DEGs	Up	Down
Leaf	21	2,132	1,032	1,100
Adventitious root	15	2,582	1,706	876

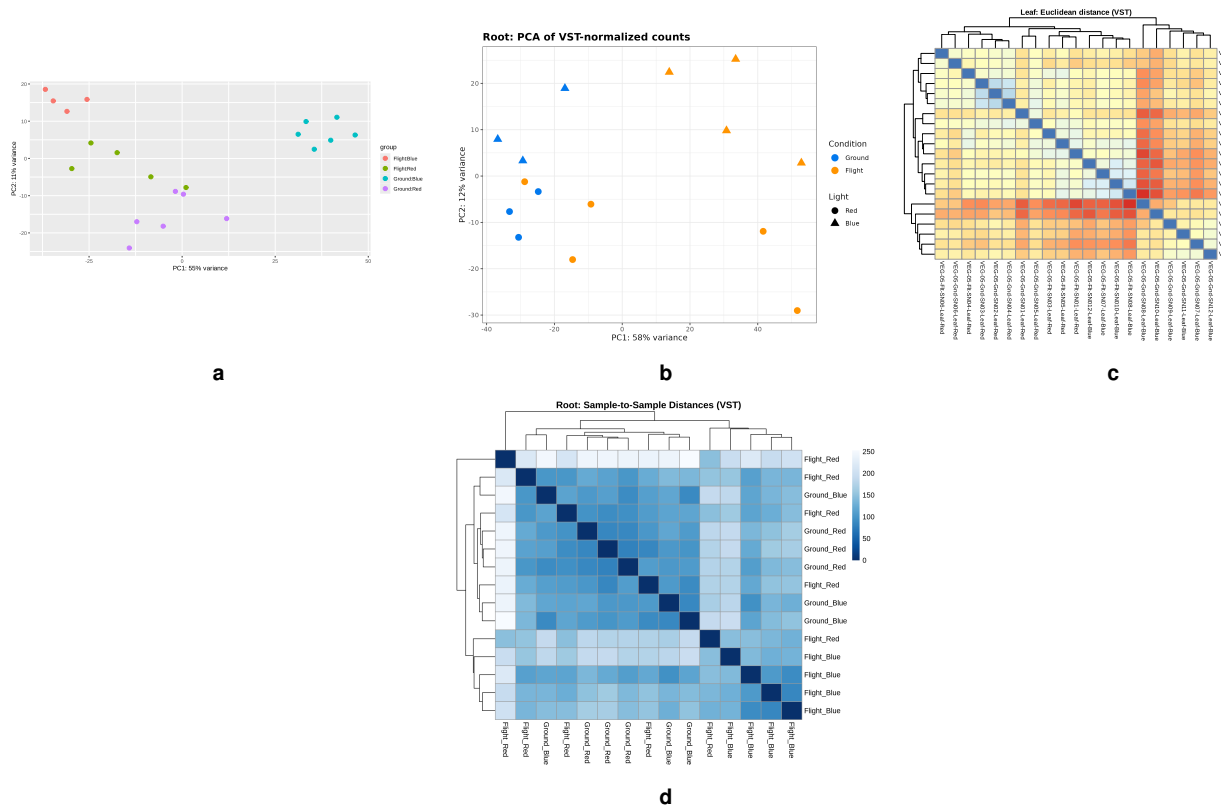


Fig. 1 | Sample quality and ordination separate spaceflight from ground controls. Principal-component analysis of variance-stabilised counts for (a) leaf and (b) adventitious-root samples, and sample-to-sample Euclidean-distance heatmaps for (c) leaf and (d) root. In both organs the dominant axis of variation separates Flight from Ground, while red- and blue-light replicates interleave within each condition, supporting the covariate design. Root samples show greater spread, consistent with the limited root ground-control replication ($n = 3$ per light condition).

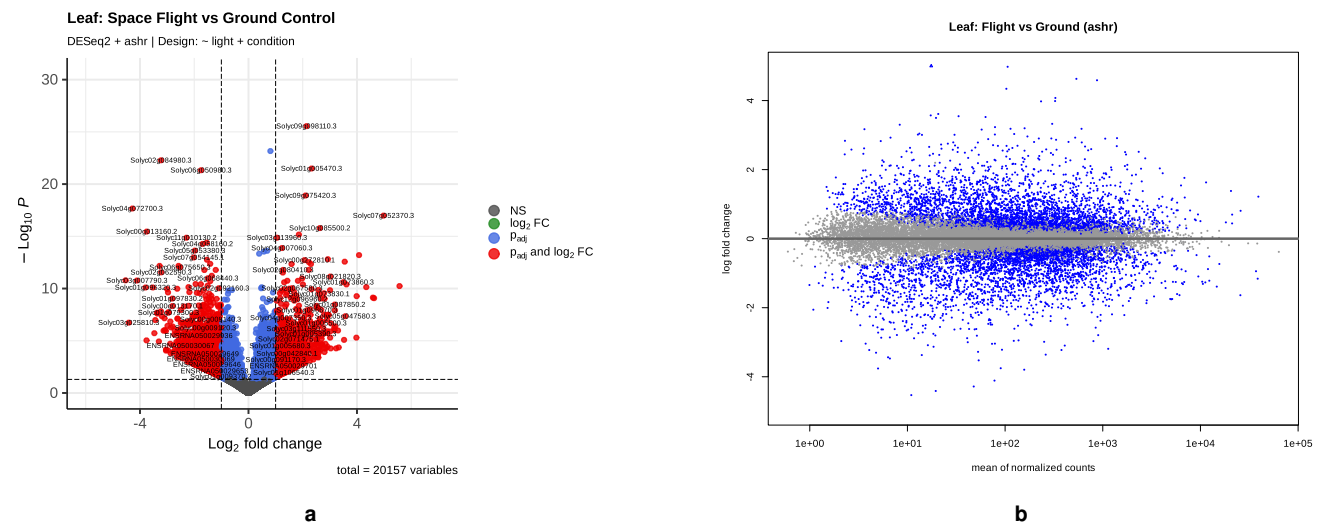


Fig. 2 | Spaceflight extensively remodels the leaf transcriptome. (a) Volcano plot and (b) MA plot of the leaf Flight-versus-Ground contrast (DESeq2, $\sim$ light + condition). Of 2,132 significant DEGs ($FDR < 0.05$, $|\log_2 FC| \geq 1$), 1,032 are up- and 1,100 down-regulated in spaceflight; effect sizes are moderate (maximum $|\log_2 FC| \approx 5.6$) and the induced/repressed distribution is broadly symmetric across the expression range.

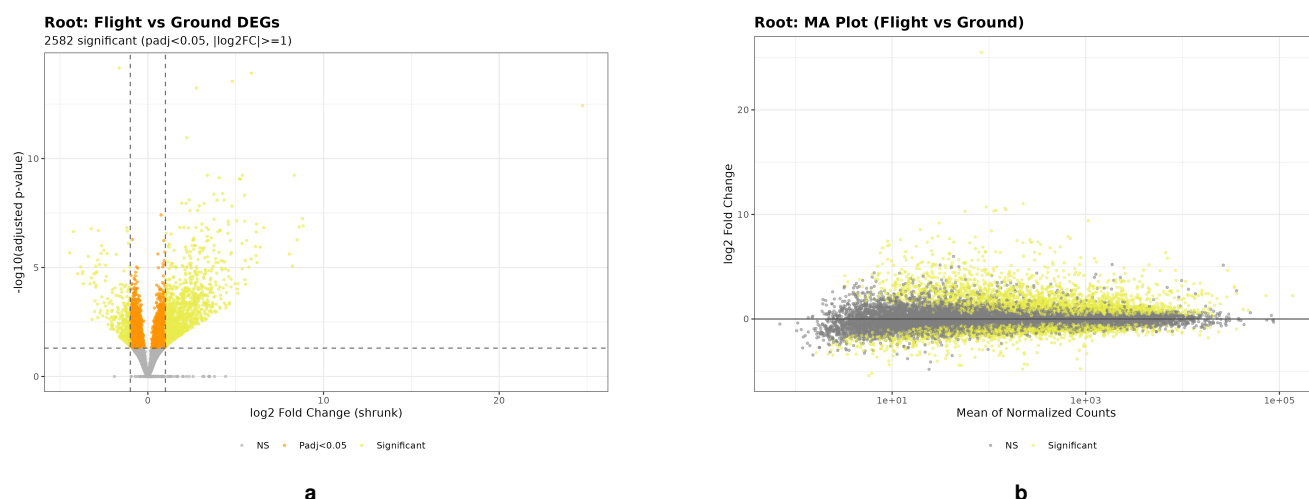


Fig. 3 | Adventitious roots show a larger, up-regulation-biased response. (a) Volcano and (b) MA plots of the root Flight-versus-Ground contrast. Of 2,582 significant DEGs, 1,706 are up- versus 876 down-regulated ($\approx 2:1$ induction bias). Root effect sizes exceed those of leaf, including switch-like induction of loci from near-zero ground baselines (strongest $\log_2$ FC $\approx +25$).

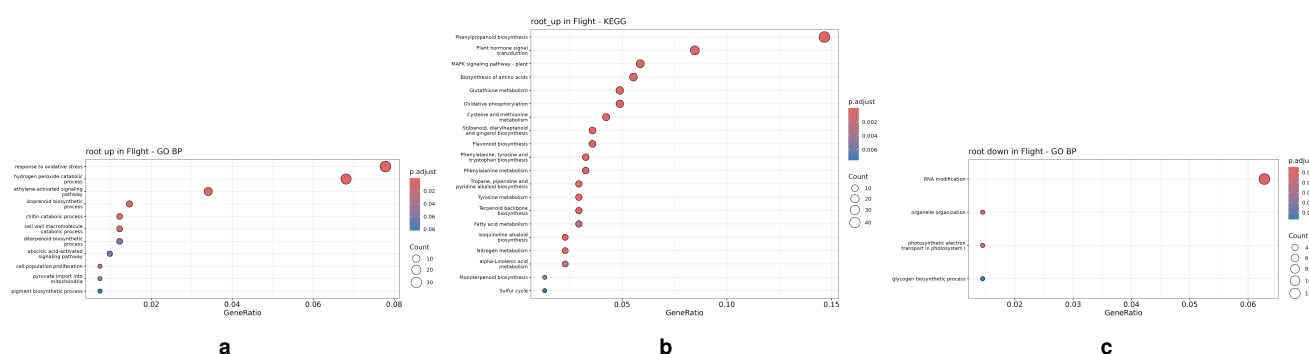


Fig. 4 | Root functional enrichment centres on oxidative-stress management and phenylpropanoid metabolism. Dot plots (dot size, gene count; colour, adjusted p ; x -axis, gene ratio) for root DEGs. (a) GO biological process, up-regulated: led by response to oxidative stress and hydrogen-peroxide catabolism. (b) KEGG, up-regulated: dominated by phenylpropanoid biosynthesis (FDR $\approx 4 \times 10^{-30}$), with flavonoid/stilbenoid biosynthesis, glutathione metabolism and plant-hormone signalling. (c) GO biological process, down-regulated.

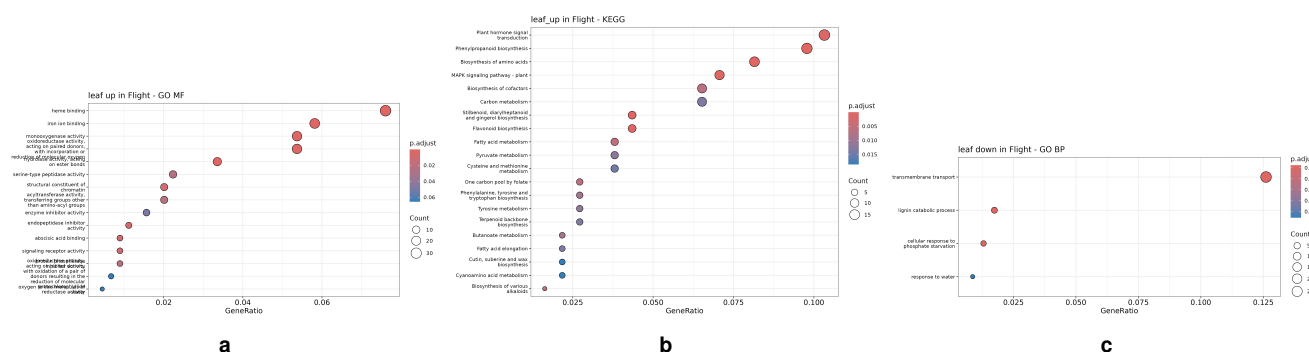


Fig. 5 | Leaf functional enrichment mirrors the root stress programme. Dot plots for leaf DEGs. (a) GO molecular function, up-regulated: heme binding, monooxygenase/oxidoreductase and iron-ion binding. (b) KEGG, up-regulated: phenylpropanoid and flavonoid biosynthesis, amino-acid biosynthesis, MAPK signalling and plant-hormone signal transduction. (c) GO biological process, down-regulated, including transmembrane transport; the plant circadian-rhythm KEGG pathway is down-regulated in both organs.

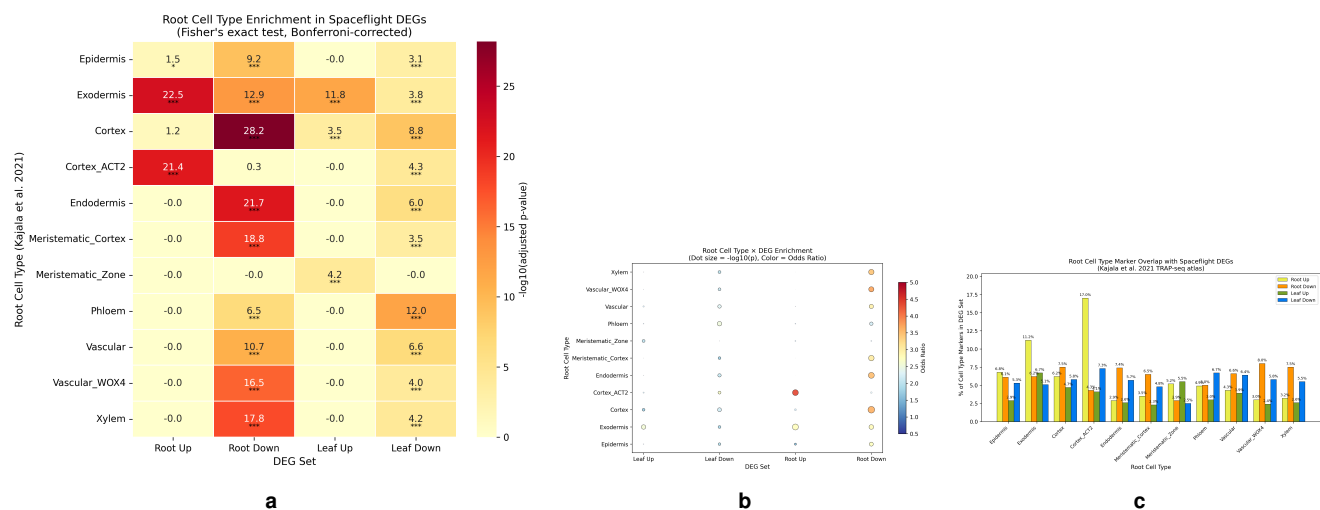


Fig. 6 | Spaceflight root DEGs localise to ground-tissue and vascular cell types. Root DEGs cross-referenced against published tomato root cell-type markers³ (Fisher's exact test, Bonferroni-corrected). **(a)** Heatmap of $-\log_{10}$ adjusted p for each cell type (rows) $\times$ DEG set (columns); cortex is most enriched among down-regulated genes ($\text{FDR} \approx 7 \times 10^{-29}$) and exodermis/Cortex_ACT2 among up-regulated genes. **(b)** Dot plot and **(c)** overlap bar plot summarising the same associations.